

Soil Health & Crop Recommendation Report

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Farmer Information

Name: ramesh
Phone: 8088178117
Location: pune
Land Size: 100.0 acres

Soil & Weather Data

N: 93.0 mg/kg
P: 87.0 mg/kg
K: 76.0 mg/kg
PH: 6.50
Temperature: 28.3
Humidity: 68.4
Rainfall: 550.0
Soil_type: alluvial soil

Recommended Crop

Predicted Crop: Rice
Model Confidence: 100.0%

Alternative Crop Options

- 2. Watermelon - Confidence: 0.0%
- 3. Banana - Confidence: 0.0%

Important Warnings

■■ Very high phosphorus levels detected. May interfere with micronutrient uptake.

Soil Treatment Suggestions

- Soil pH (6.5) is within optimal range (5.5-7.0) for rice.
- Nitrogen (93.0 mg/kg) is within optimal range for rice.
- Phosphorus (87.0 mg/kg) is excessive for rice. Optimal range: 30-50. High P can interfere with micronutrient uptake.
- Potassium (76.0 mg/kg) is excessive for rice. Optimal range: 30-50. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

- Soil NPK levels are already optimal for rice.
Maintain soil health with organic compost and crop rotation.
- **Organic Alternatives:**
 - Compost/FYM: 5-10 tons/acre (provides N, P, K)
 - Green Manure: Leguminous crops (fixes N)
 - Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

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Crop-Specific Tips for Rice

Ensure consistent water supply and transplant in puddled field. Maintain 5-7 cm water depth.

Additional Notes

- These recommendations are based on soil analysis and ML predictions.
- Actual results may vary based on local conditions and farming practices.
- Consult with local agricultural experts before making major decisions.
- Regular soil testing is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for soil health.